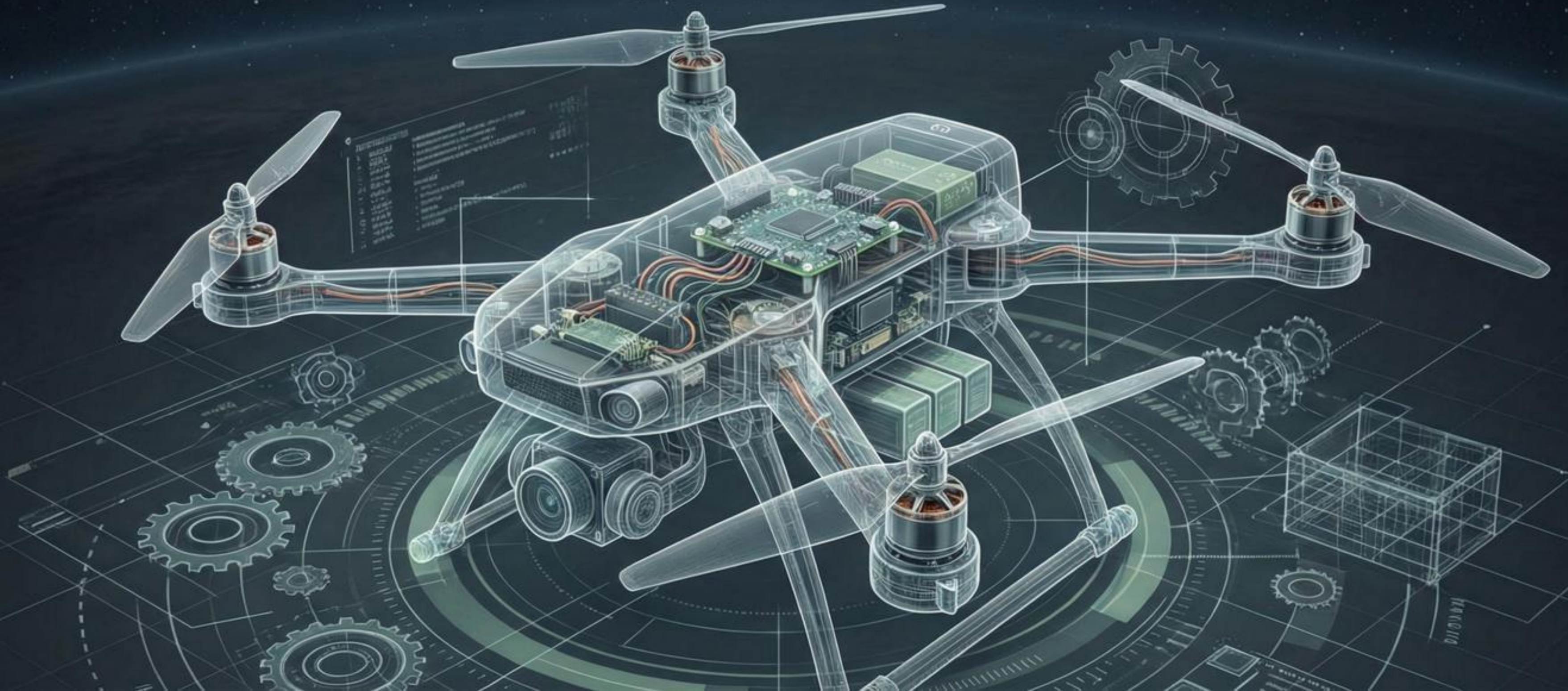


DRONE ENGINEERING DEEP DIVE

Prepared by Ahamed Jazif



DRONE DESIGN METRICS: QUANTIFYING DECISIONS

THRUST-TO-WEIGHT RATIO (TWR)



Ratio of total thrust produced to drone's total weight.

Higher TWR means better vertical acceleration & payload, but requires powerful, heavier motors & batteries, impacting endurance.

POWER LOADING (W/kg)



Power consumed per unit of drone weight.

Lower power loading means higher efficiency & endurance, often using larger, slower-spinning propellers and high-capacity batteries, reducing agility.

DISK LOADING (kg/m²)



Weight supported per unit of total rotor disk area.

Lower disk loading leads to better efficiency & endurance (larger props), but results in larger frame size, more susceptibility to wind, and reduced agility.

ENDURANCE (Time)



Maximum flight time under specific conditions.

Driven by battery capacity, aerodynamic efficiency, and power consumption. Maximizing endurance often compromises agility, payload, and speed.

AGILITY INDEX (Relative)



Measure of responsiveness and maneuverability.

Higher agility requires high TWR, low rotational inertia, and responsive motors, usually at the cost of stability, efficiency, and endurance.

Understanding these metrics allows for informed component selection and design optimization, moving beyond guesswork to data-driven engineering decisions.

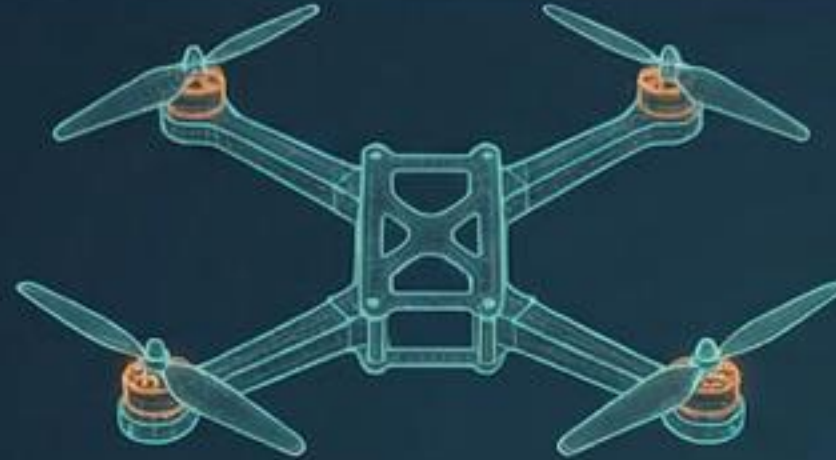
DRONE ENGINEERING DEEP DIVE

ARM LAYOUT & STRUCTURAL ANALYSIS



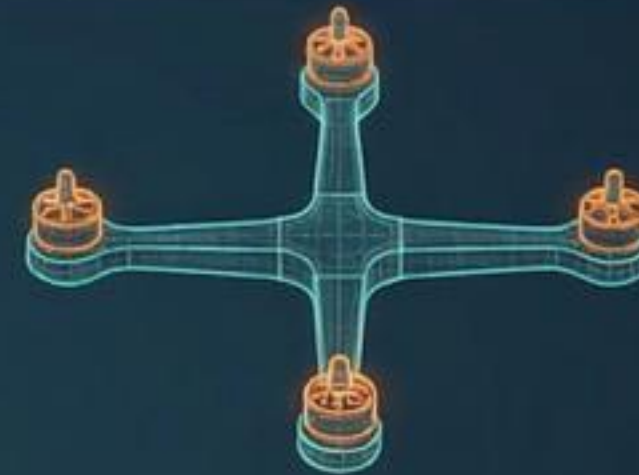
X LAYOUT

Symmetrical, balanced thrust, good maneuverability, common for racing & aerial photography.



H LAYOUT

Clear camera view, easy to mount payloads, often used for filming, less agile than X.



PLUS (+) LAYOUT

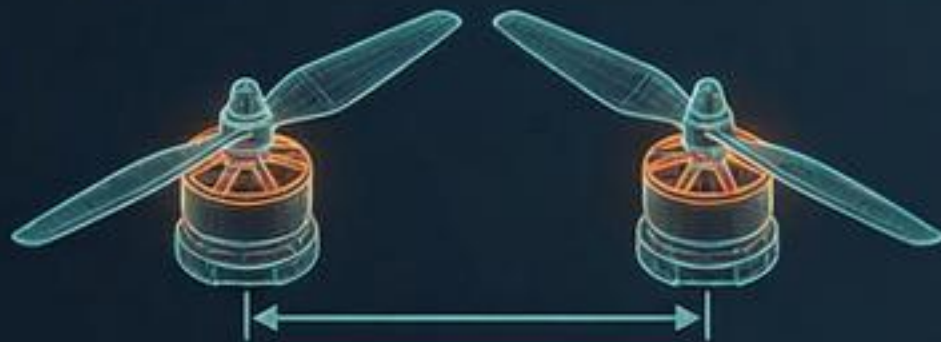
Simple control, intuitive for beginners, larger profile, reduced maneuverability.



SQUARE LAYOUT

Stable platform, large payload area, less common, more drag.

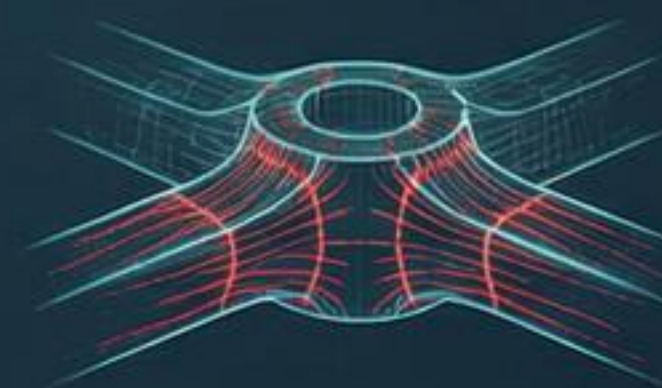
MOTOR SPACING DETERMINES MAX PROP DIAMETER & YAW AUTHORITY



Larger Spacing = Larger Props = More Thrust/Efficiency, But Lower Yaw Response.

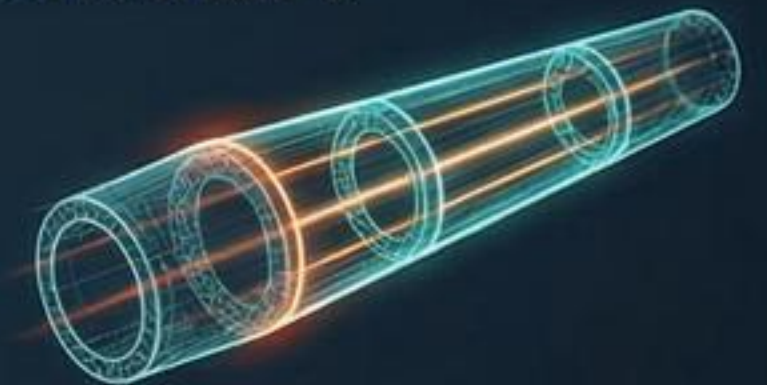


Smaller Spacing = Smaller Props = Higher Yaw Authority (Faster Rotation), But Less Efficiency.



STRESS CONCENTRATION AT ARM-HUB JOINTS

High forces at connection points, requires reinforcement, potential failure point.



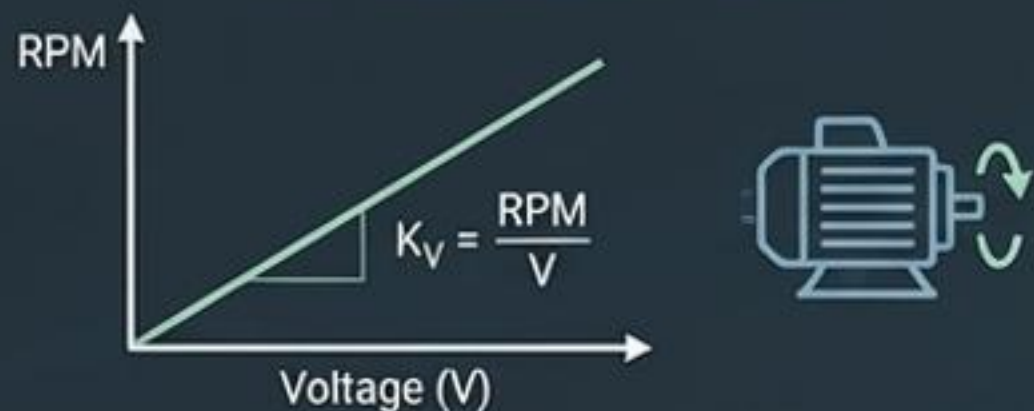
TAPERED SECTIONS & MATERIAL OPTIMIZATION

Reduces mass at extremities while maintaining stiffness near hub, increases efficiency.

Motor Parameters & Flight Envelope Analysis

FUNDAMENTAL MOTOR CONSTANTS

KV Constant (RPM/V)



No-Load Current (I_0)



Current drawn by the motor when spinning freely at a specific voltage with zero load.
Represents mechanical losses.

Internal Resistance (R_m)



Electrical resistance of the motor windings.
Causes power loss (heat) proportional to the square of current ($I^2 R_m$).

THRUST TABLE ANALYSIS & SAFE FLIGHT ENVELOPE

Reading Thrust Tables

Input Voltage	Throttle Signal	Current Draw	Thrust (g)	Power Consumption	Efficiency Metric	Propeller Speed
Voltage (V)	Throttle (%)	Current (A)	Thrust (g)	Power (W)	Efficiency (g/W)	RPM
2.3	0	0.1	100	8.7	0.93	1200
2.3	10	1.8	200	13.5	0.90	1300
2.6	20	2.3	300	18.5	0.74	1450
2.8	50% Throttle	3.3	400	20.8	0.98	1800
2.4	50	4.1	600	17.3	0.95	1800
2.8	60	4.7	800	21.1	0.80	2000
2.6	70	5.6	1000	31.8	0.92	2040

Deriving Hover Throttle



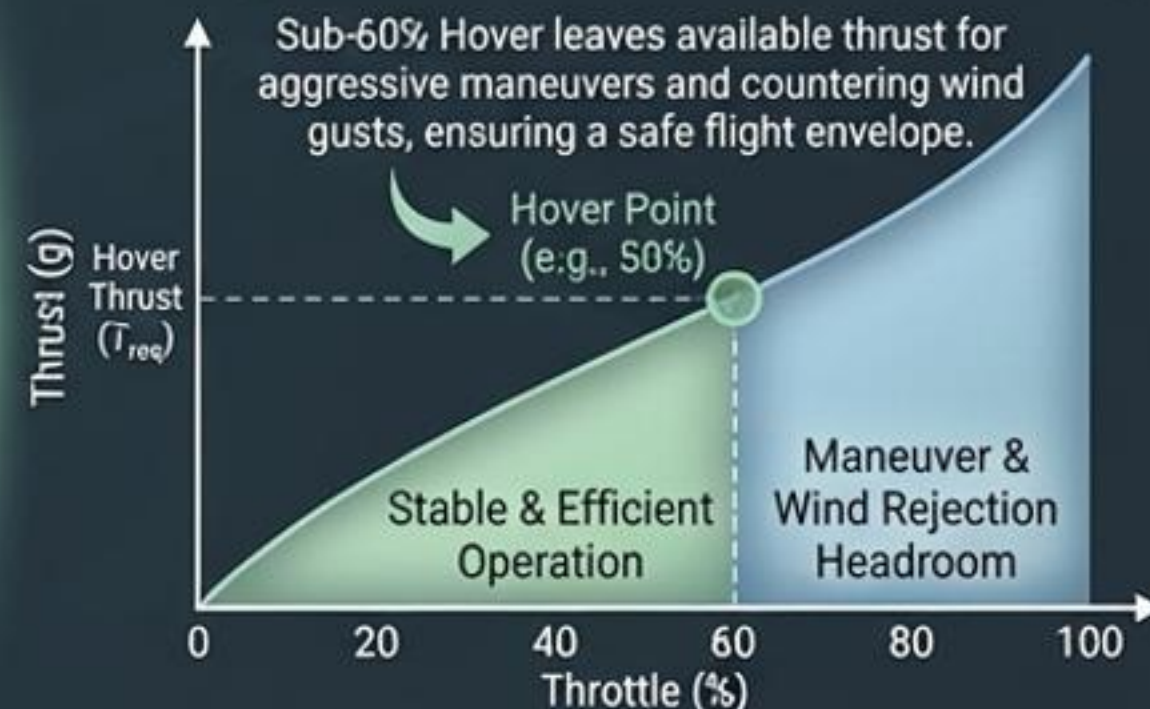
Total Weight (TW)
= Drone Weight + Payload

Required Thrust per Motor (T_{req}) =
$$\frac{TW}{\text{Number of Motors}}$$

Locate T_{req} in Thrust Table
→ Find corresponding Throttle %.

e.g., $TW = 1600g$, 4 Motors → $T_{req} = 400g/\text{motor}$.
From table, 400g corresponds to ~48% Throttle.

Safe Flight Envelope (Sub-60% Operation)



PROPELLER AERODYNAMICS: DESIGN PARAMETERS & PERFORMANCE

1. Fundamental Propeller Parameters



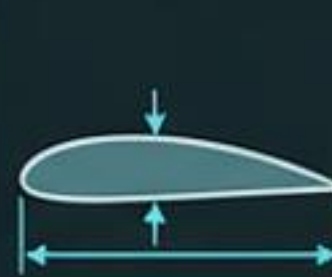
DIAMETER

The diameter of the propeller disc. Larger diameter generally provides more thrust and efficiency for a given power, but limits responsiveness.



PITCH

The theoretical distance a propeller advances forward in one revolution. Higher pitch means more speed but requires more torque.



CHORD

The width of the propeller blade. Affects the amount of lift generated and drag produced.



REYNOLDS REGIME

Dimensionless number describing fluid flow. Drones typically operate in low Reynolds numbers, affecting boundary layer and lift/drag characteristics.

$$Re = \frac{\rho * V * L}{\mu}$$

2. Linking Theory to Thrust

STATIC THRUST EQUATION

$$T = C_T * \rho * n^2 * D^4$$

Simplified equation relating thrust (T) to thrust coefficient (C_T), density (ρ), RPM (n), and diameter (D).

BLADE ELEMENT THEORY (BET)



Calculates total thrust and torque by integrating the forces acting on small, individual blade elements along the span.

PERFORMANCE PREDICTION

Provides a more detailed and accurate understanding of propeller performance across different flight conditions compared to static models.

3. Propeller Selection Trade-off: 5030 vs. 6040

5030 PROPELLER (5.0" Diameter, 3.0" Pitch)



Responsive,
Higher Top Speed,
Lower Torque Required



Trade-off: Lower Efficiency at Hover,
Higher Noise



RESPONSIVENESS
& SPEED



EFFICIENCY
& HOVER

Engineers must balance this trade-off based on mission requirements (e.g., racing vs. aerial photography).

6040 PROPELLER (6.0" Diameter, 4.0" Pitch)



Efficient Hover,
More Lifting Capacity,
Smoother Flight

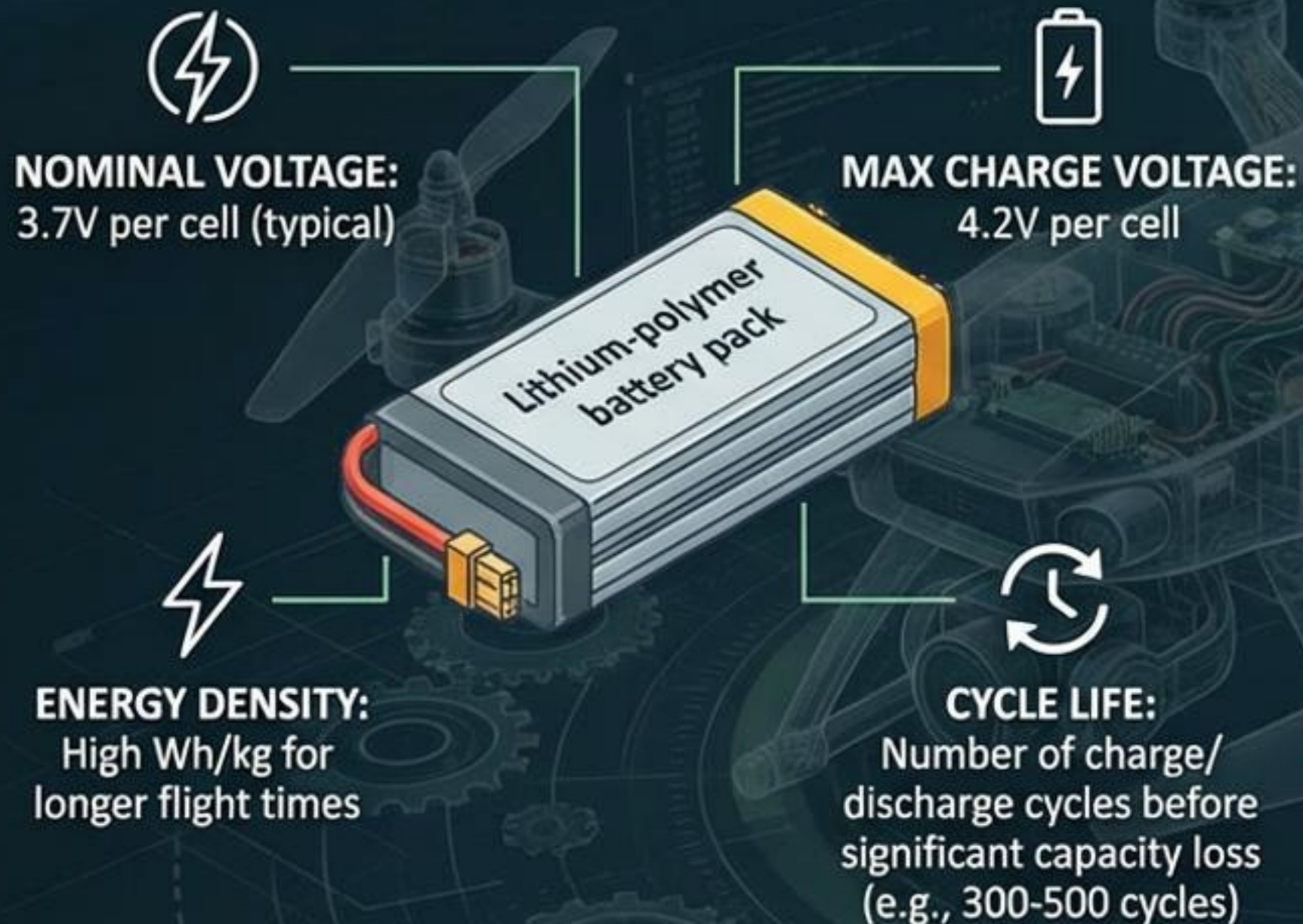


Trade-off: Lower Responsiveness, Lower Top Speed, Higher Torque Required

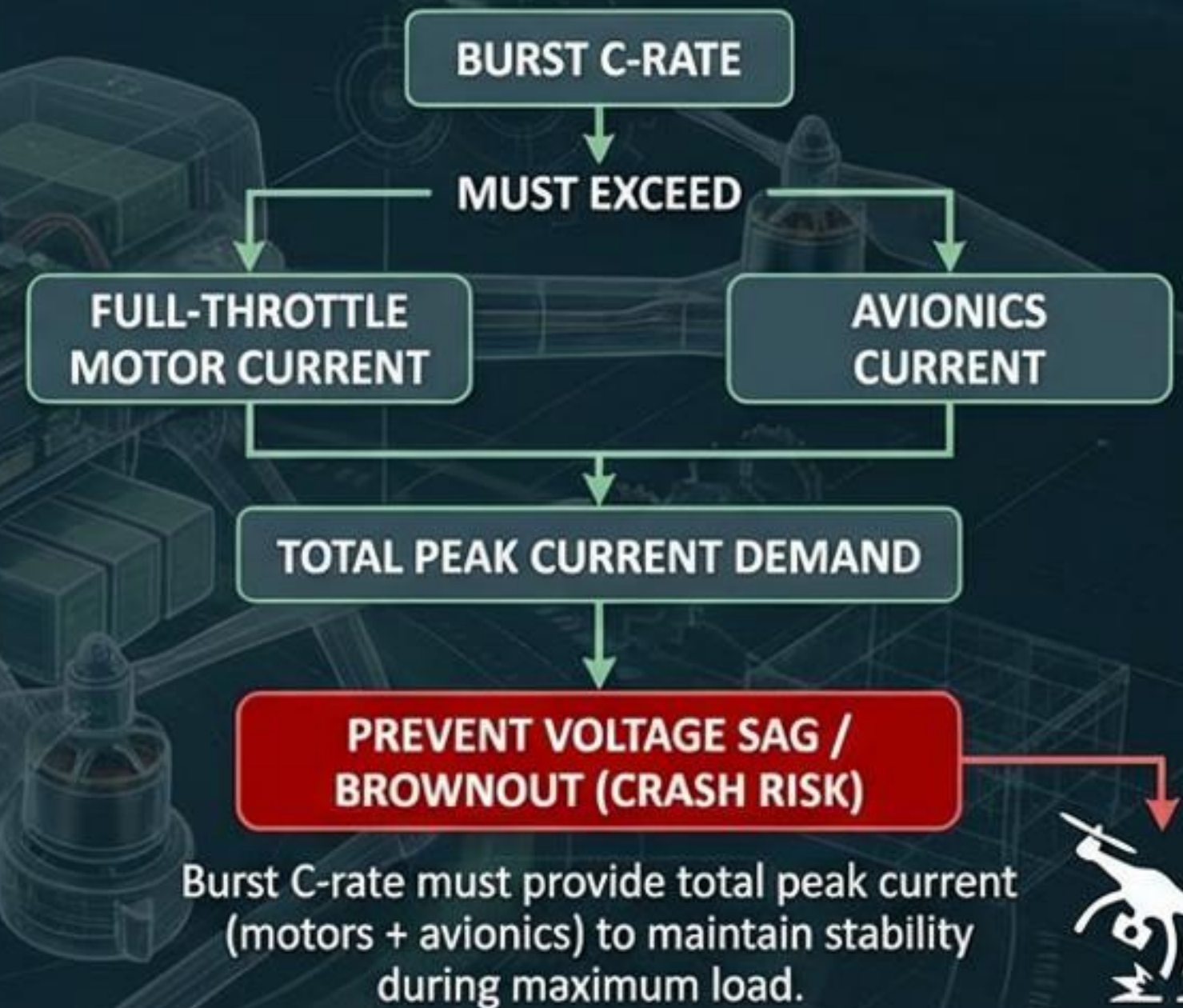
DRONE BATTERY SPECIFICATIONS & PERFORMANCE

Prepared by Ahamed Jazif

BATTERY FUNDAMENTALS



C-RATE & POWER DELIVERY



POWER DISTRIBUTION BOARD (PDB) & SIGNAL INTEGRITY



COPPER WEIGHT & TRACE TEMP RISE

Thick copper layers (e.g., 2oz, 3oz) handle high currents with minimal resistance, reducing heat generation and voltage drop under load.

ELECTRONIC
SPEED CONTROLLER
(ESC)

ELECTRONIC
SPEED CONTROLLER
(ESC)

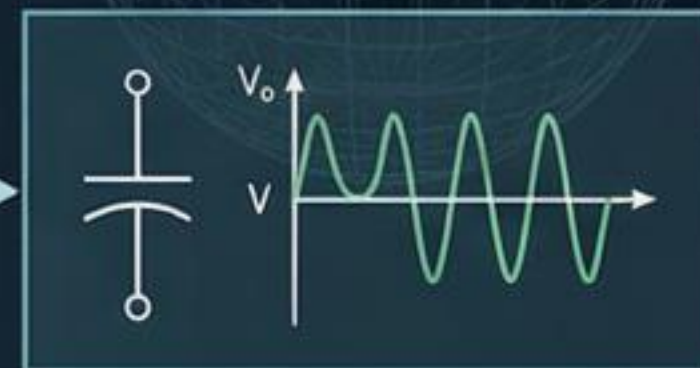
ELECTRONIC
SPEED CONTROLLER
(ESC)

LiPo
BATTERY

FLIGHT
CONTROLLER
(LOGIC)

ELECTRONIC
SPEED CONTROLLER
(ESC)

LiPo
BATTERY



LOW-ESR CAPACITORS & PWM VOLTAGE SPIKES

Low-Equivalent Series Resistance (ESR) capacitors act as rapid energy reservoirs, smoothing out high-frequency PWM voltage fluctuations from ESCs.

NOISY
POWER



PDB + CAPS
FILTERING

CLEAN POWER

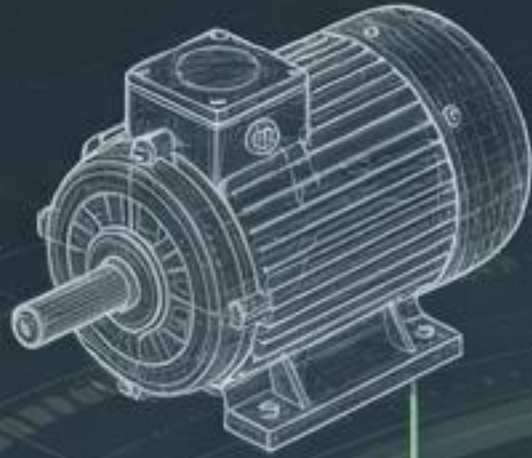
RELIABLE FLIGHT
CONTROLLER

Ensuring stable voltage and clean power supply for reliable flight controller operation, especially during peak load transients.

BATTERY ENDURANCE CALCULATION

SPREADSHEET FORMULA

$$\text{Avg. Current (A)} \times \text{Voltage (V)} = \text{Watts (W)}$$



$$\text{Wh} \div \text{W} = \text{Flight Time (Hours/Minutes)}$$

20% RESERVE MANDATE



Avoid over-discharge to protect battery health.

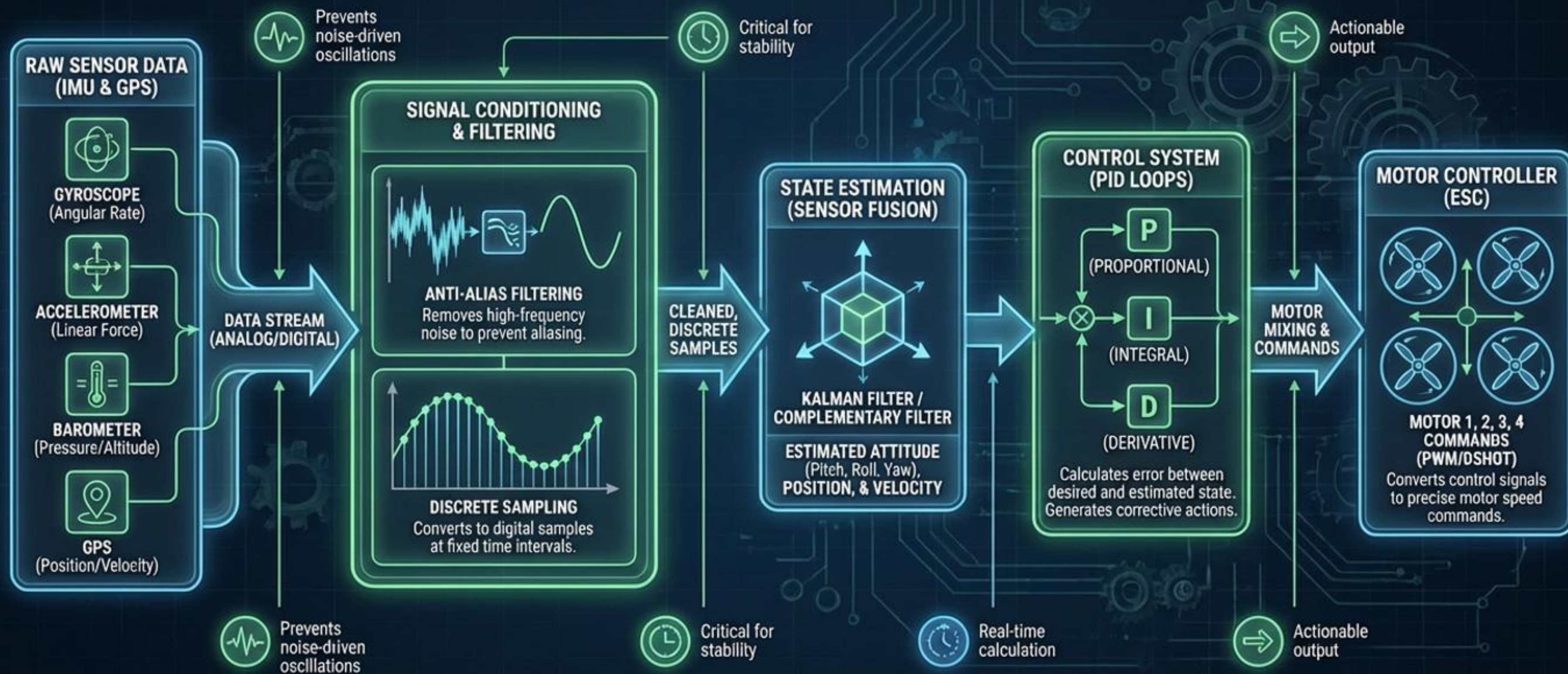


CLASSROOM TO PRACTICAL ENDURANCE

Students validate limits experimentally.

Translate physics into real-world application.

SENSOR DATA PROCESSING & CONTROL LOOP FLOW



PID CONTROLLER & TUNING WORKFLOW

Prepared by Ahamed Jazif

PID TERMS DEFINED



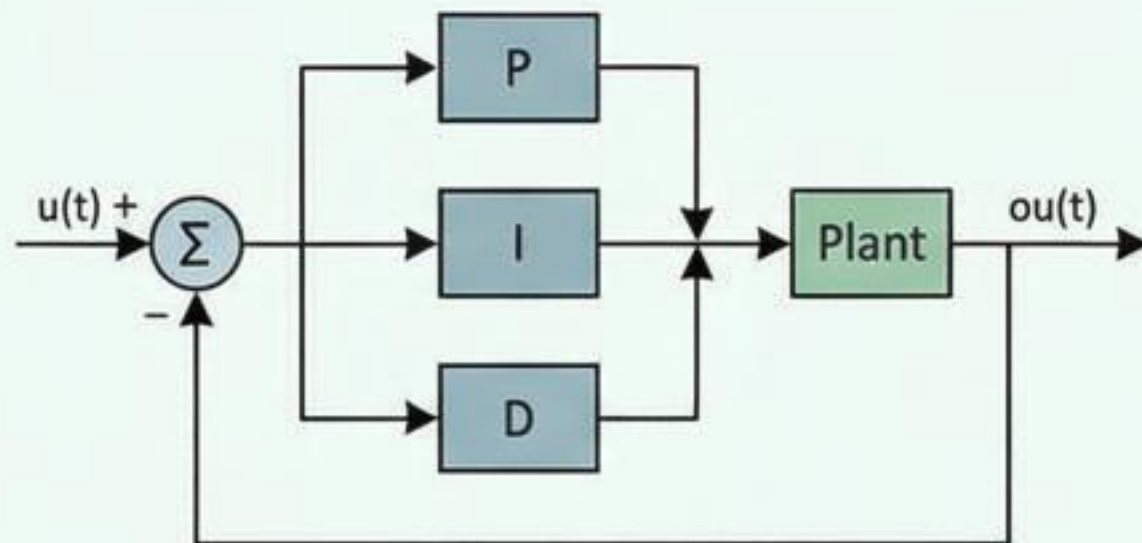
PROPORTIONAL (P): Corrects present error.



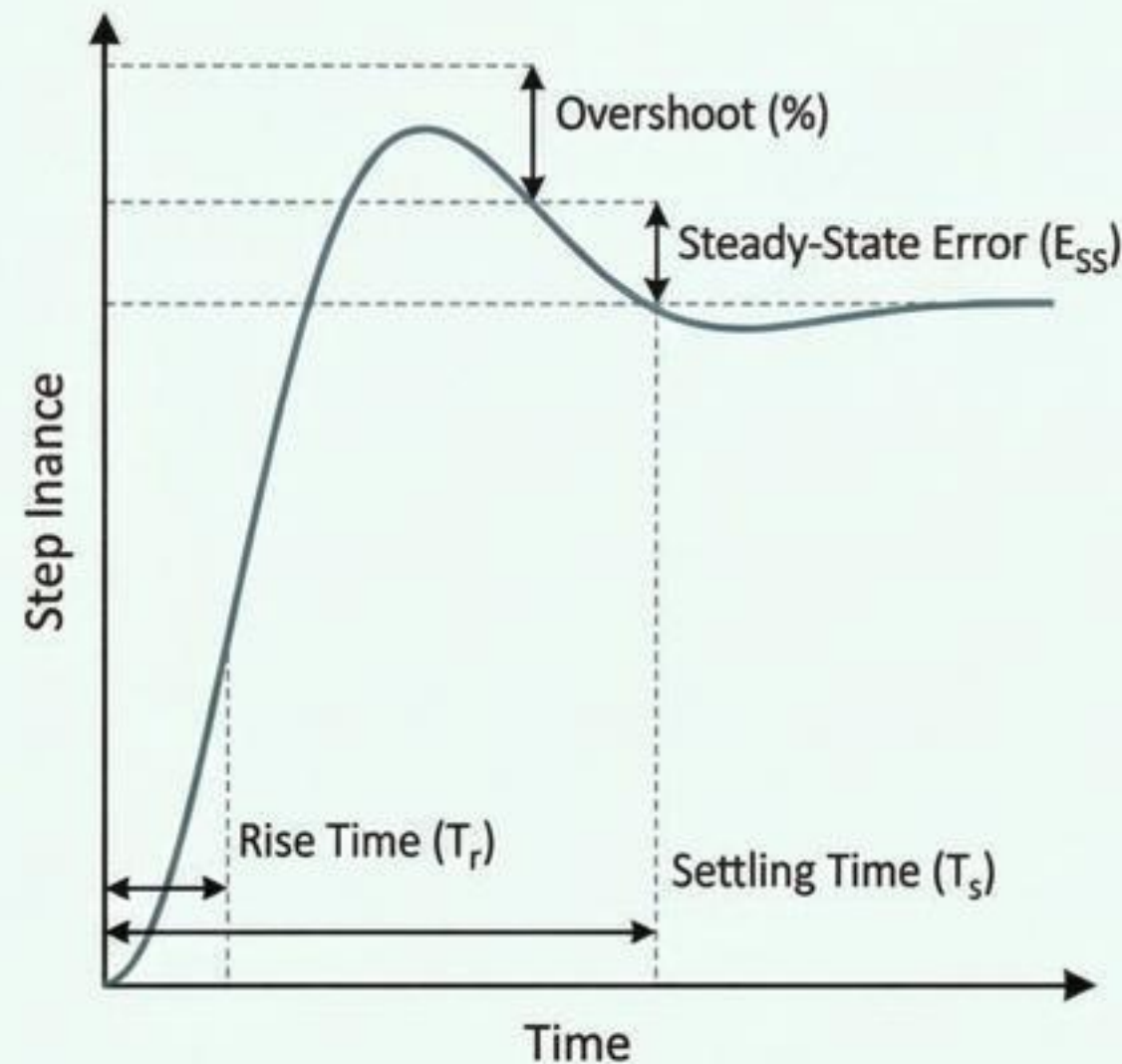
INTEGRAL (I): Corrects past cumulative error.



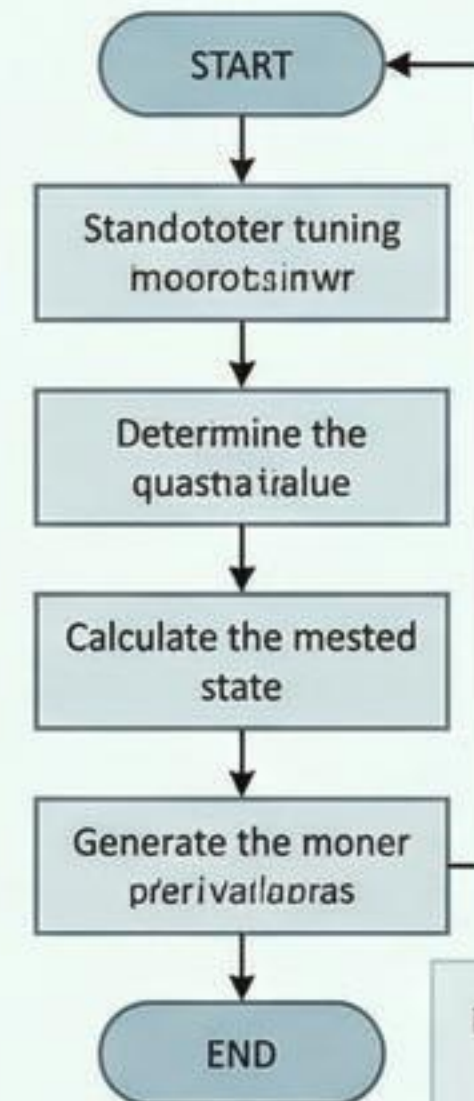
DERIVATIVE (D): Predicts future error to prevent overshoot.



PERFORMANCE METRICS (STEP-RESPONSE)



ZIEGLER-NICHOLS FOR QUADROTORS



Ultimate gain	K_p	K_i
K_u	$0.5K_u$	$0.2K_u$
T_u	0.07	0.12
$K_i = \frac{K_u}{T_u}$	$\frac{2K_p}{6T_u}$	$\frac{2K_d}{6T_u}$
$K_d = \frac{K_p}{K_u}$	$\frac{K_u}{0.03}$	$\frac{K_{li}}{0.03}$

REPEATABLE WORKFLOW FOR STABLE, CRISP FLIGHT PERFORMANCE.

SENSOR CALIBRATION & FLIGHT STABILITY: CONNECTING MATH TO MOTION

CONSEQUENCES OF INACCURATE CALIBRATION



DRIFT (ACCEL/GYRO ERROR)
Causes slow position deviation due to incorrect attitude or acceleration estimates.

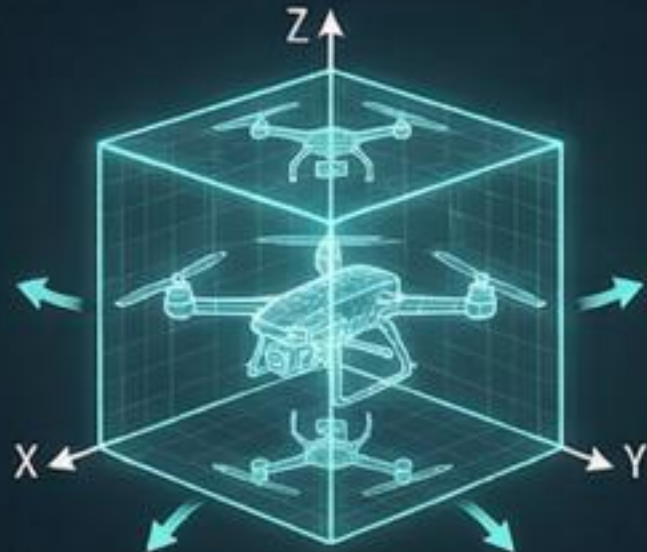


TOILET BOWLING (MAG ERROR)
Causes spiraling in Loiter as the drone misinterprets its heading relative to position corrections.



THE CALIBRATION PROCESS & MATH MODELS

SIX-POSITION ACCEL CAL



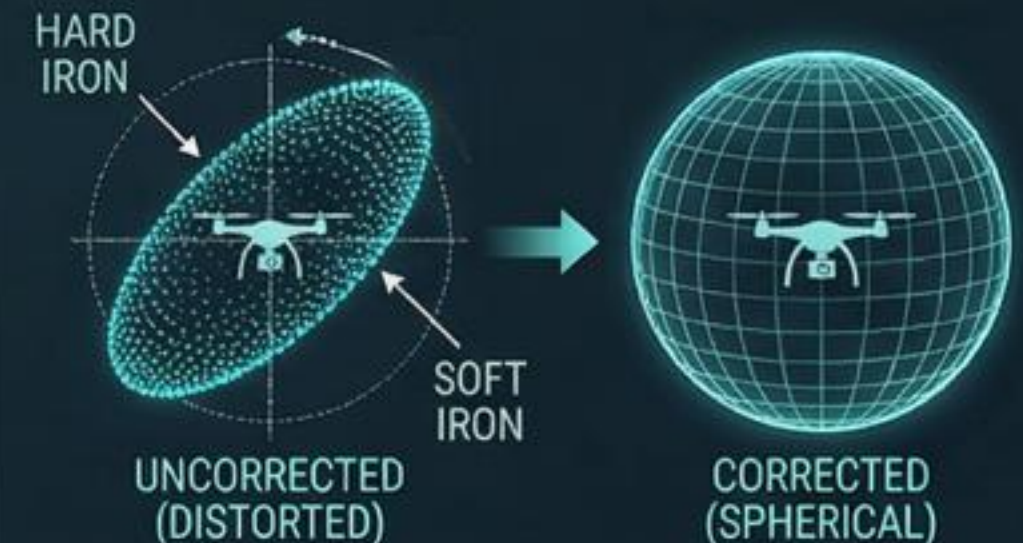
Estimates sensor offsets & scaling by finding gravity vector in 6 static orientations. Corrects tilt errors.

GYRO BIAS ESTIMATION



Calculates and subtracts static zero-rate output (bias) when drone is still. Prevents drift from bias integration.

MAG HARD/SOFT IRON COMP



Compensates for permanent (hard iron) and induced (soft iron) magnetic field distortions. Ensures accurate heading.

LINKING MATH TO MOTION: Accurate calibration ensures the mathematical models (bias, scaling, distortion matrices) align with physical reality, preventing the specific observable instabilities like drift and toilet bowling.

DRONE COMMUNICATION PROTOCOLS & FAIL-SAFE SYSTEMS

Prepared by Ahamed Jazif

COMMUNICATION PROTOCOLS

PROTOCOL	FRAME PERIOD	LATENCY	DESCRIPTION
PWM (Pulse Width Modulation)	100ms (variable)	High (>20ms)	Legacy, analog, one wire per channel.
SBUS (Serial Bus)	14ms (typical)	Low (<10ms)	Digital, inverted serial, multiple channels on one wire.
CRSF (Crossfire)	2.5ms (typical)	Ultra-Low (<5ms)	Modern digital, high update rate, low latency, telemetry.

Frame Period: Time between signal packets.

Latency: Delay from input to action.

RECEIVER FAIL-SAFE MECHANISMS



THROTTLE DROP REQUIREMENT: Throttle must drop below a preset pulse (e.g., 950µs) after signal loss to prevent runaway and initiate fail-safe actions.

WHY: Ensures safety by disarming or activating automated return in case of control link failure.

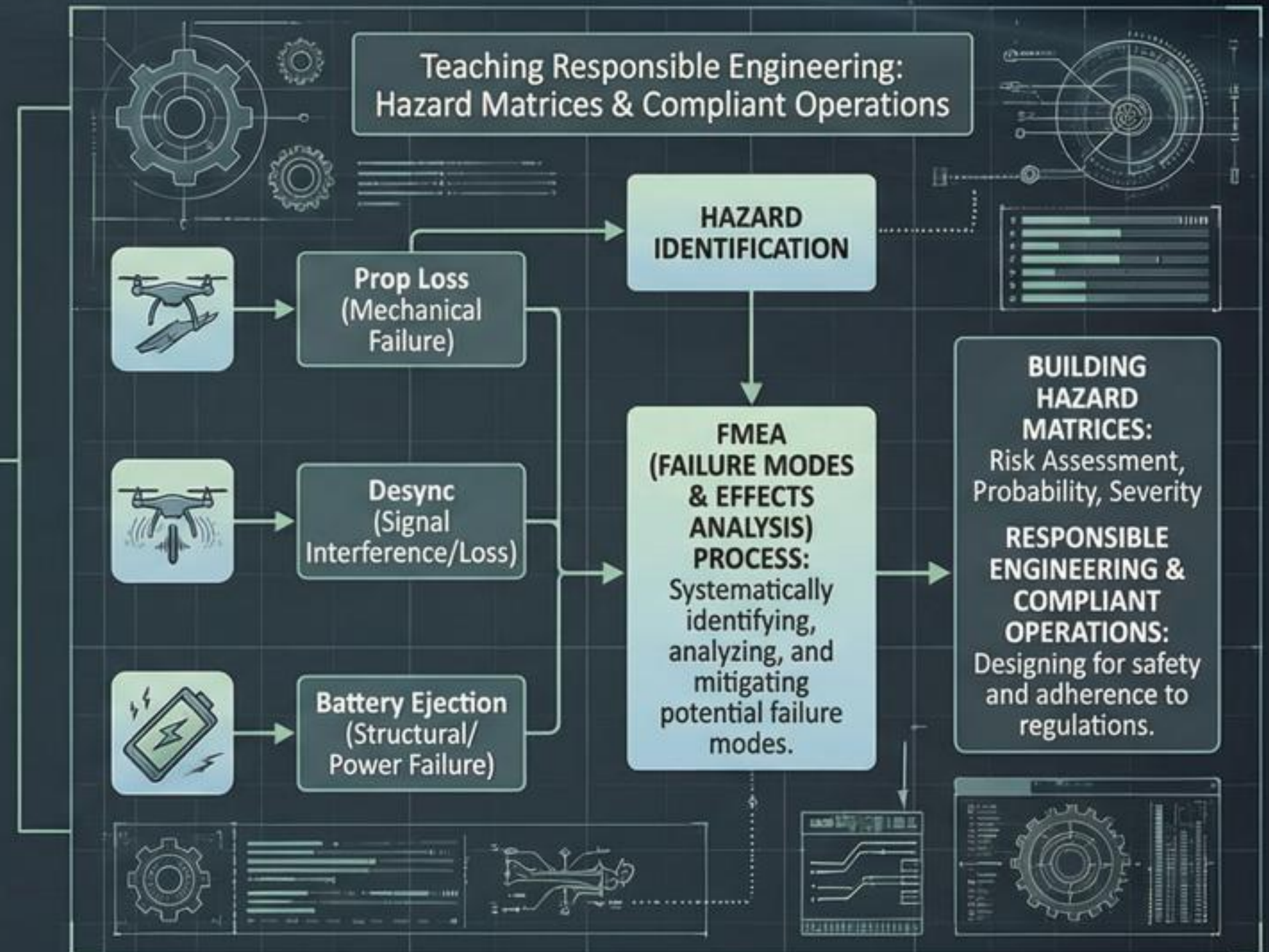
DRONE REGULATIONS & ENGINEERING RESPONSIBILITY

Prepared by Ahamed Jazif

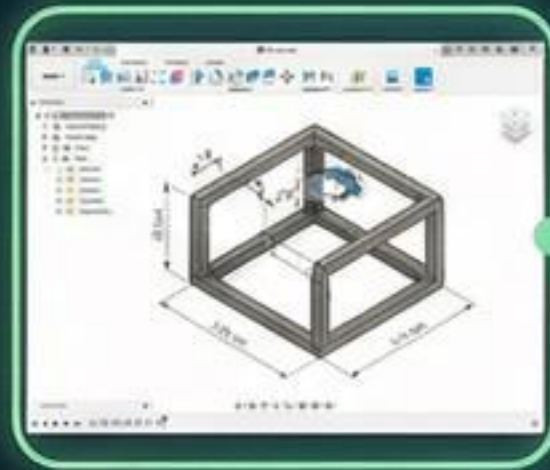
CORE DRONE REGULATIONS (Summary)



ENGINEERING RESPONSIBILITY & FMEA PROCESS



DESIGN & MANUFACTURING WORKFLOW



**PARAMETRIC
FRAME MODELING
(FUSION 360)**



STL EXPORT



**SLICER ORIENTATION
(FOR STRENGTH)**



**ITERATIVE
FIT CHECKS**

• EMPHASIZE DFM & TRACEABILITY •



DFM TOLERANCES

Ensure
manufacturability
within tolerances.



SUPPORT REMOVAL

Minimize
post-processing
and waste.



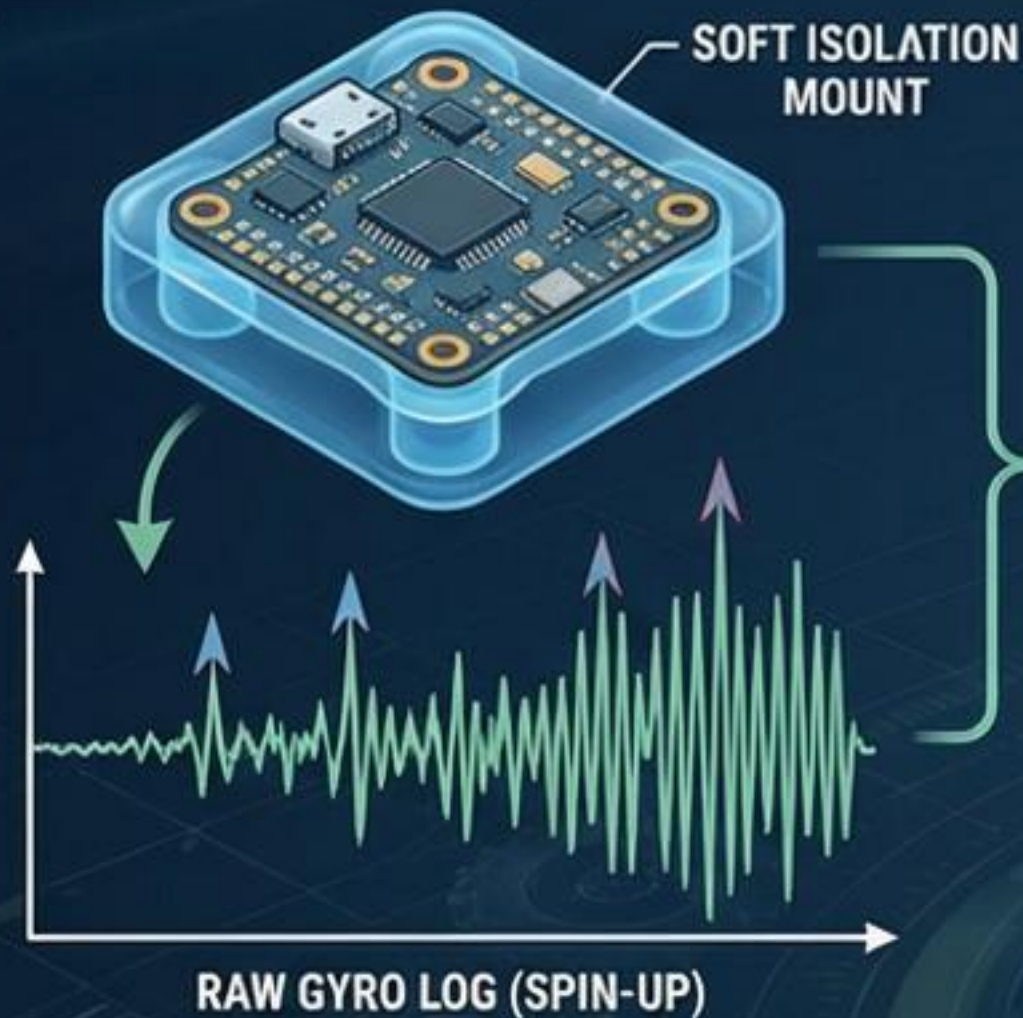
LIVE WEIGHT TRACKING

Keep design goals
traceable.

VIBRATION ANALYSIS & MITIGATION

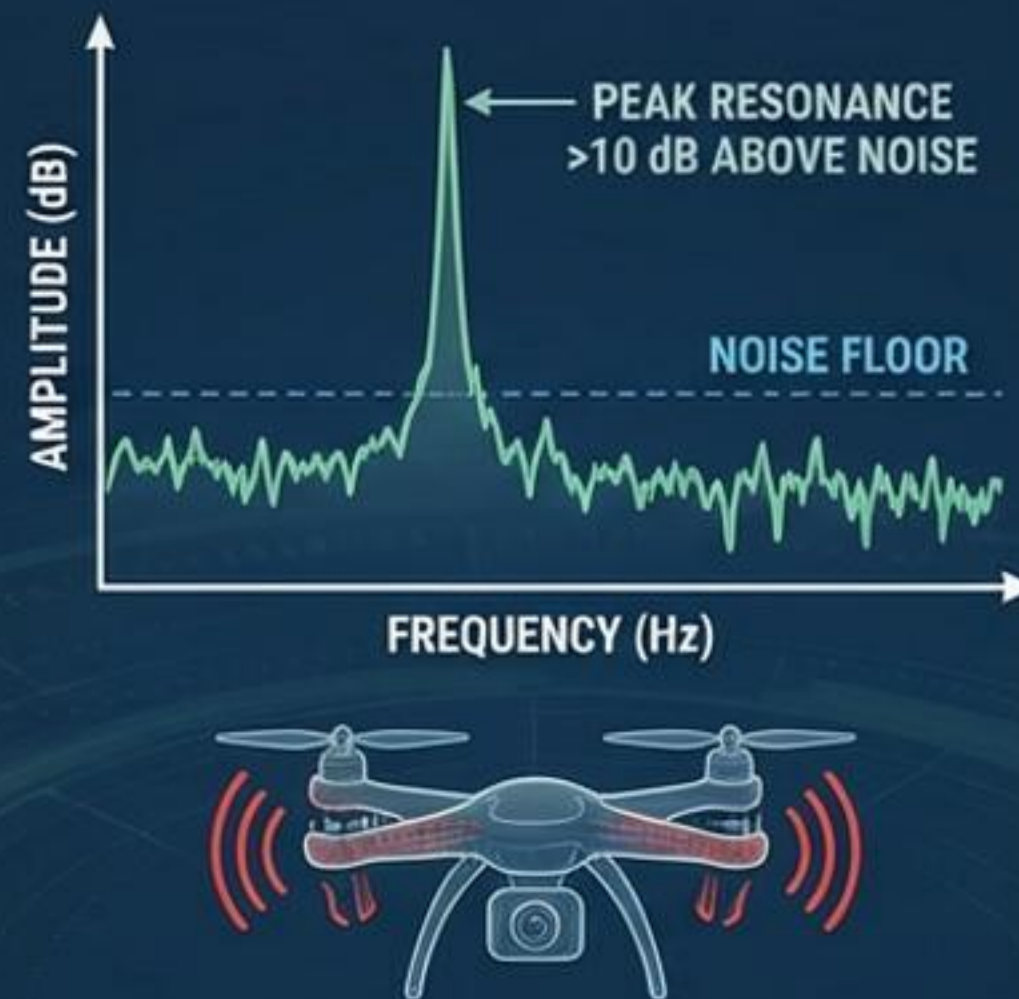
Prepared by Ahamed Jazif

1. DIAGNOSIS: DATA COLLECTION



Mount FC on soft isolation. Log raw gyro data during propeller spin-up tests.

2. ANALYSIS: FFT & RESONANCE



Analyze FFT peaks. Identify if resonance is >10 dB above noise floor.

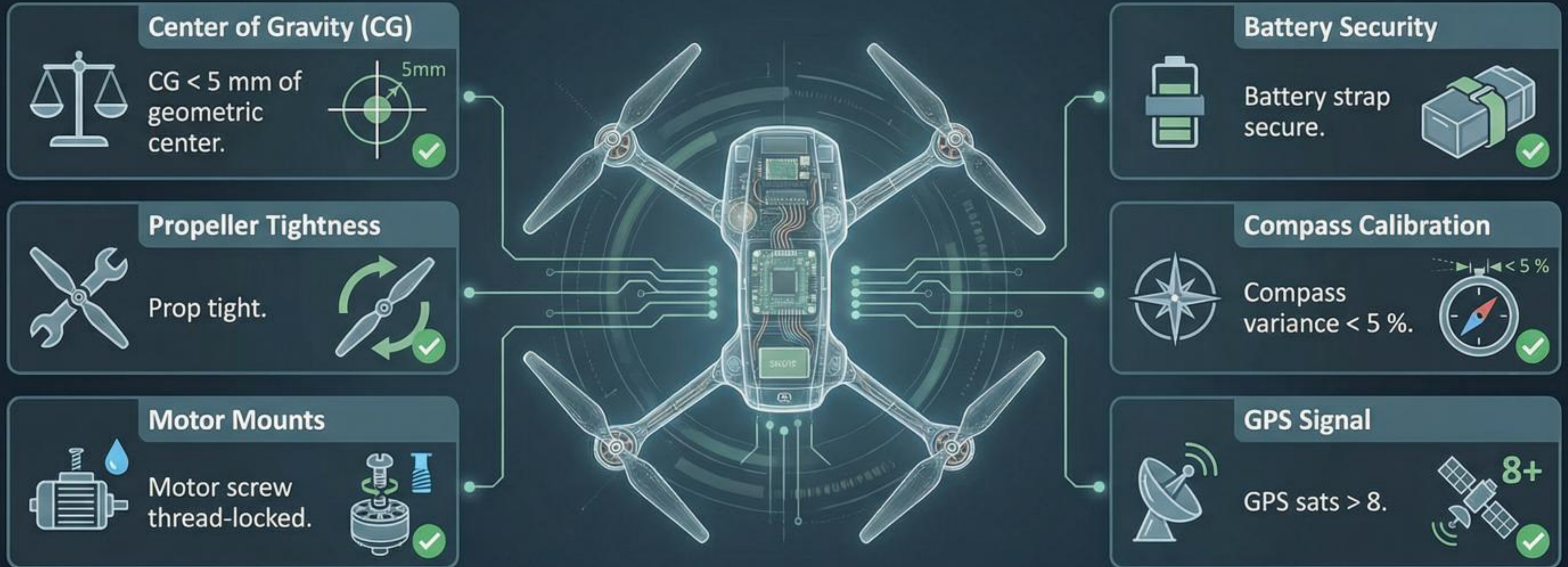
3. RESOLUTION: ELIMINATION



Redesign arm structure for increased stiffness or add dampers to eliminate oscillations, preventing video compromise and instability.

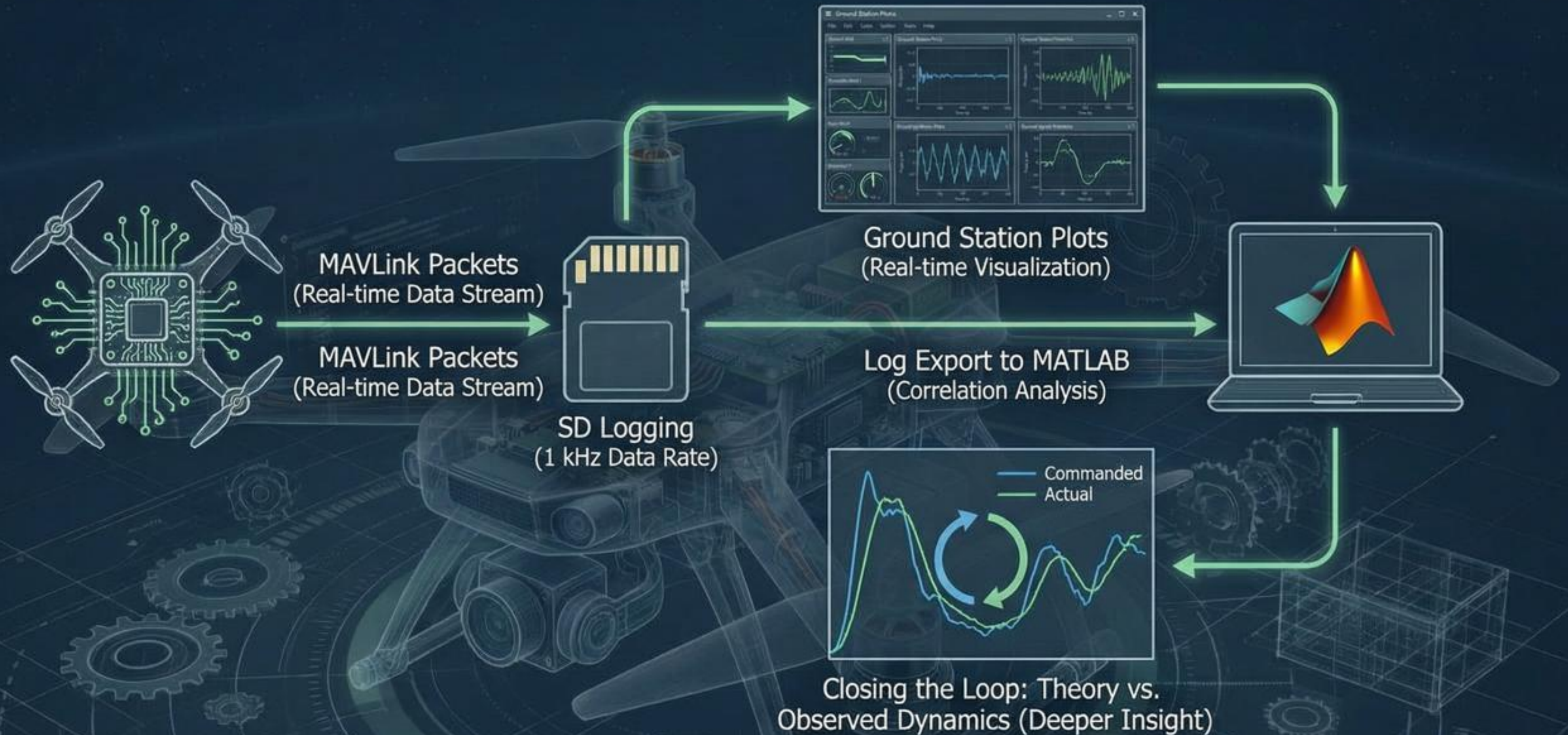
PRE-FLIGHT CHECKLIST

Prepared by Ahamed Jazif



ENSURING REPEATABLE SAFE FLIGHTS & STUDENT CONFIDENCE.

MAVLink, SD Logging, and Data Analysis Loop



SYSTEMS THINKING: ITERATIVE DESIGN RE-EVALUATION

Evaluating the impact of advanced modules on drone performance.

